

Secure your own WI-FI Network

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Task ID: Task 3

Organization: Future Interns

Tools Used:

- ❖ Nmap (for network scanning)
- ❖ Wireshark (for packet analysis)
- ❖ Windows Command Line
- ❖ Router Web Interface

1. Identify Devices on the Network:

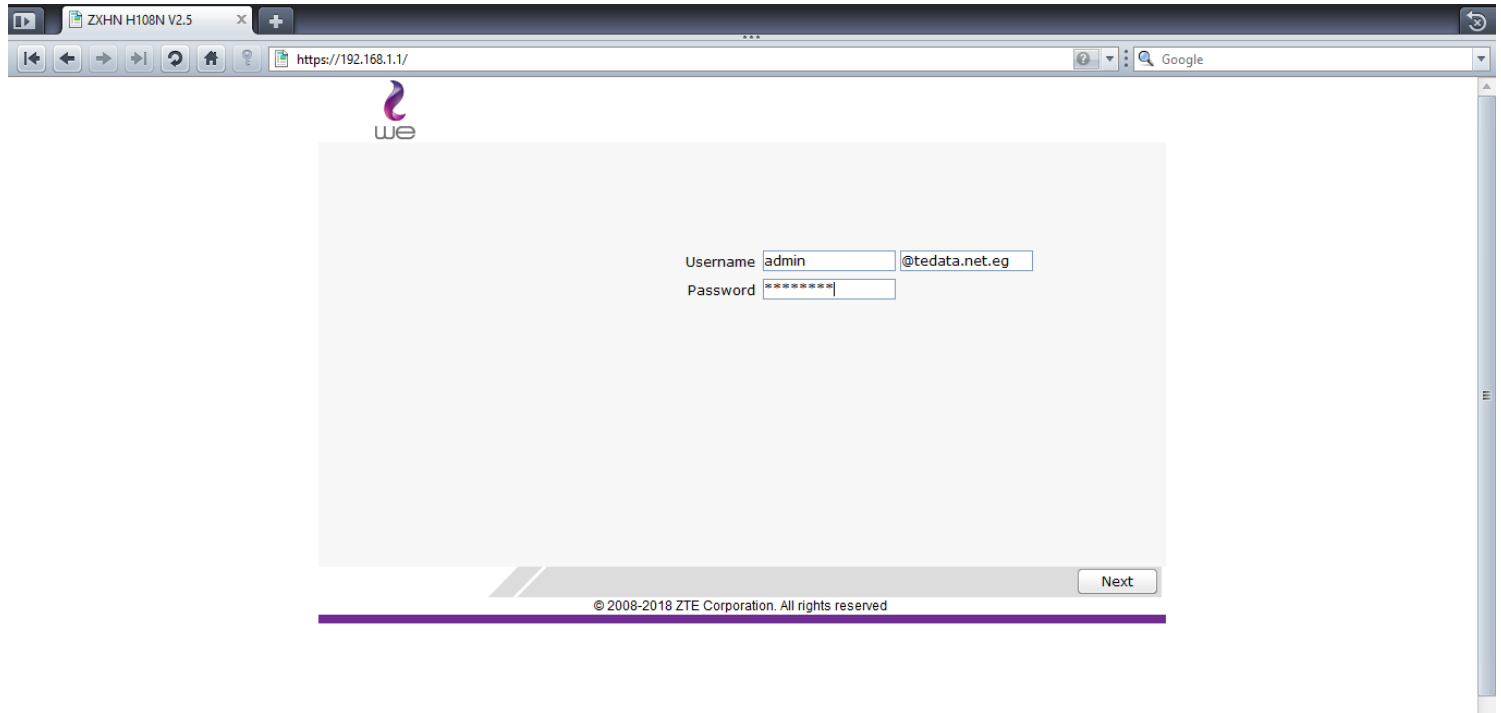
- **Scanned the local network using** → `nmap -sV -sS -O 192.168.1.0/24`

Findings:

- Three active IP addresses were discovered on the network:
 - 192.168.1.1: Router (ZTE ZKHN H108N V2.5).
 - 192.168.1.2: realme-5 phone.
 - 192.168.1.3: Samsung Galaxy Tab A 2016.
 - 192.168.1.4: Windows 10 device.
-
- Devices included a router, a tablet, a Mobile Phone, and a Windows PC, consistent with expected network usage.

2. Access Router Admin Panel:

- Router accessed via browser at `https://192.168.1.1`
- Logged in using admin credentials (default or user-defined)



Findings:

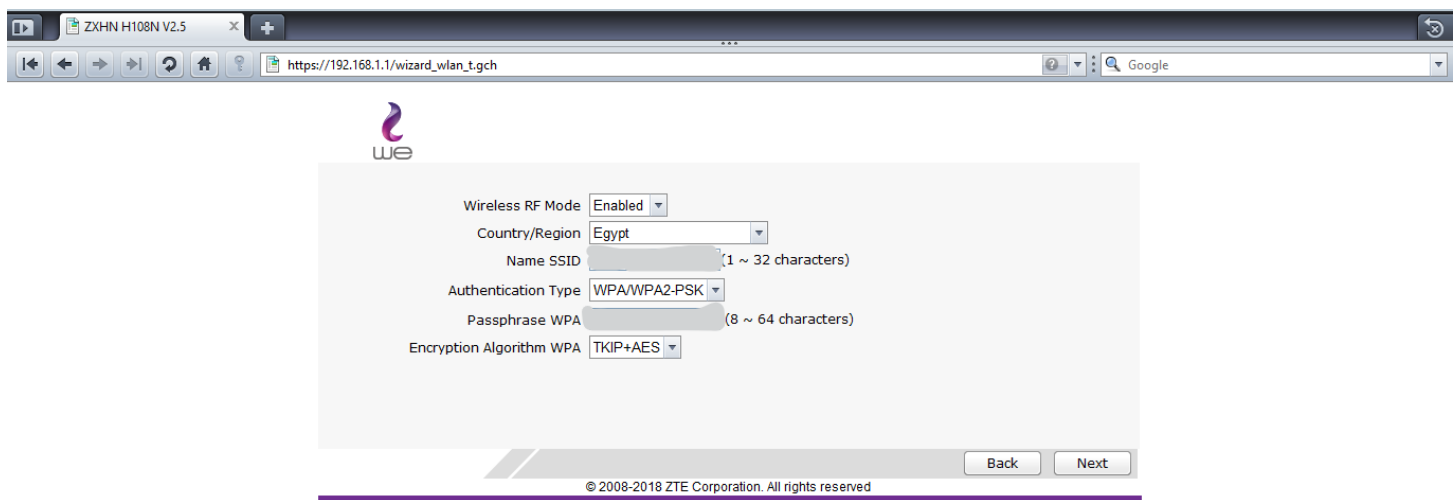
- Tabs visible: Wireless RF Mode, Country/Region, Name (SSID), Authentication Type, Passphrase, Encryption Algorithm.

Settings reviewed:

- Wireless RF Mode: Enabled.
- Country/Region: Egypt.
- Authentication Type: WPA/WPA2-PSK.
- Encryption Algorithm: WPA TKIP+AES.
- Passphrase: 8–64 characters (redacted).

Analysis:

- The admin panel is accessible over HTTPs, which is secure.
- WPA/WPA2-PSK allows both WPA and WPA2 protocols. WPA with TKIP is outdated and vulnerable to attacks like KRACK (CVE-2017-13077). WPA2 is more secure, but the router should enforce WPA2-only or WPA3.
- TKIP+AES encryption allows both TKIP and AES. TKIP is deprecated and insecure (e.g., vulnerable to MIC key recovery attacks). AES-only should be used.
- The copyright notice (© 2008-2018 ZTE Corporation) suggests outdated firmware, consistent with the Nmap-detected Linux kernel 2.4, which is highly vulnerable.
- Confirmed ability to see connected devices and settings via the WLAN tab.



3. Port Scan on Internal Devices:

Tool Used: Nmap

Nmap is a powerful network scanning tool used to detect hosts, services, and potential vulnerabilities on a network. The `-sV` flag is used to detect service versions running on open ports, and `-sS` performs a stealth SYN scan.

Target IPs:

- 192.168.1.1
- 192.168.1.2
- 192.168.1.3
- 192.168.1.4

Device 192.168.1.1 (Router/Gateway):

The IP address 192.168.1.1 is the router or default gateway in the local network, acting as the primary bridge between internal devices and the internet.

```
C:\Users\dell>nmap -sV -sS -O 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-02 15:06 Egypt Daylight Time
Nmap scan report for 192.168.1.1
Host is up (0.0051s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Mini web server 1.0 (ZTE ZXV10 W300 ADSL router http config)
443/tcp    open  ssl/https    Mini web server 1.0 ZTE corp 2005.
52869/tcp  open  upnp         Portable SDK for UPnP devices 1.6.6 (UPnP 1.0)
MAC Address: 88:8E:88:88:88:88 (zte)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: OSs: Linux 2.4.17, Linux; Device: broadband router; CPE: cpe:/h:zte:zxv10_w300, cpe:/o:montavista:linux_kernel:2.4.17, cpe:/o:linux:linux_kernel
```

Open Ports:

- **80/tcp (HTTP):** Mini web server 1.0 (ZTE ZXV10 W300 ADSL router http config).
- **443/tcp (HTTPS):** Mini web server 1.0 (ZTE corp 2005).
- **52869/tcp (UPnP):** Portable SDK for UPnP devices 1.6.6 .

Analysis:

- **Port 80 (HTTP):** Open HTTP access to the admin panel is a significant vulnerability. If not redirected to HTTPS, credentials can be intercepted in plain text.
- **Port 443 (HTTPS):** Secure access is available, but HTTP should be disabled to enforce HTTPS-only.
- **Port 52869 (UPnP):** UPnP is enabled, which can be exploited for port forwarding or DDoS attacks (e.g., UPnP-based DDoS amplification). It should be disabled unless required.
- **OS Version:** The router runs an outdated Linux kernel (2.4.17 or 2.4.6), vulnerable to known exploits (e.g., CVE-2003-0462). Firmware updates are critical.

Device 192.168.1.2 (Realme-5 mobile phone):

```
Nmap scan report for realme-5 (192.168.1.2)
Host is up (0.036s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
5060/tcp   filtered  sip
MAC Address: [REDACTED] (Unknown)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop
```

Analysis:

- No open ports were detected; all 1000 scanned ports are closed (reset).
- This indicates a strong firewall or no running services, making the device (likely a Samsung Galaxy Tab A 2016) secure against external attacks.
- The closed ports returning TCP RST (reset) confirm the device is active but not accepting connections.

Device 192.168.1.3 (Samsung Galaxy Tab A 2016):

```
Nmap scan report for galaxy-tab-a-2016 (192.168.1.3)
Host is up (0.020s latency).
All 1000 scanned ports on galaxy-tab-a-2016 (192.168.1.3) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: [REDACTED] (Samsung Electronics)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop
```

Analysis:

- No open ports were detected; all 1000 scanned ports are closed (reset).
- This indicates a strong firewall or no running services, making the device secure against external attacks.
- The closed ports returning TCP RST (reset) confirm the device is active but not accepting connections.

Device 192.168.1.4 (Windows Device):

```
Nmap scan report for 192.168.1.4
Host is up (0.00066s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
902/tcp    open  ssl/vmware-auth  VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp    open  vmware-auth      VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
16992/tcp  open  http             Intel Active Management Technology User Notification Service httpd 11.8.97.4739
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1809 - 21H2
Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:11.8.97.4739
```

Open Ports:

- **135/tcp (MSRPC):** Microsoft Windows RPC netbios-ssn.
- **445/tcp (Microsoft-DS):** Microsoft Windows.
- **902/tcp (VMware-auth):** VMware Authentication Daemon 1.10 (Uses VNC, SOAP).
- **912/tcp (VMware-auth):** VMware Authentication Daemon 1.0 (Uses VNC, SOAP).
- **19992/tcp (HTTP):** Intel Active Management Technology User Notification Service httpd 11.8.97.4739

Analysis:

- **Port 135 (MSRPC):** RPC is often targeted for exploits like EternalBlue (CVE-2017-0144), which led to WannaCry ransomware. It should be firewalled or disabled.
- **Port 445 (Microsoft-DS):** SMB is vulnerable to exploits like SMBv1 attacks (e.g., CVE-2017-0144). This port should be disabled or restricted.
- **Ports 902 and 912 (VMware-auth):** Used by VMware for authentication. If VMware is not in use, these ports should be closed to prevent unauthorized access (e.g., brute-force on VNC).
- **Port 19992 (Intel AMT):** Intel AMT can be exploited if misconfigured (e.g., CVE-2017-5689, remote privilege escalation). It should be disabled unless required.

- **OS Version:** Windows 10 1809–21H2 is supported and Updated to the latest version in May,2025

General Analysis:

IP Address	Ports Open	State	Likely Device	Comments
192.168.1.1	80 (HTTP), 443 (HTTPS), 52869 (UPnP)	Open	Router (ZTE ZKHN H108N V2.5)	HTTP and UPnP pose vulnerabilities; outdated firmware detected.
192.168.1.2	None	Closed	Realme-5 Mobile Phone	Highly secured, no services running.
192.168.1.3	None	Closed	Samsung Galaxy Tab A 2016	Highly secured, no services running.
192.168.1.4	135 (MSRPC), 445 (Microsoft-DS), 902/912 (VMware-auth), 19992 (HTTP)	Open	Windows 10 PC	High-risk ports (SMB, RPC, AMT) detected; potential for exploitation.

Final Interpretation:

- The router has significant vulnerabilities due to HTTP and UPnP.
- The Windows device (192.168.1.4) has multiple high-risk ports open.
- The Galaxy Tab (192.168.1.2) is secure with no open ports.

4. Analyze Wi-Fi Traffic (Passive Mode)

Tool Used: Wireshark

Capture Details:

- Interface used: Intel(R) Dual Band Wireless-AC 8265
- Number of packets captured: 3000+
- Source and Destination IPs/Ports analyzed

Capturing from Intel(R) Dual Band Wireless-AC 8265 #2 - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
185	36.538538	192.168.1.1	239.255.255.250	SSDP	NOTIFY * HTTP/1.1
186	38.200507	157.240.252.60	192.168.1.4	TLSv1.2	Application Data
187	38.200603	192.168.1.4	157.240.252.60	TCP	34904 > https [ACK] Seq=1 Ack=47 win=1023 Len=0
188	40.030036	192.168.1.4	192.168.1.1	DNS	Standard query A crl3.digicert.com
189	40.055402	192.168.1.1	192.168.1.4	DNS	Standard query response CNAME crl3.edge.digicert.com CNAME cac-ocsp.digicert.com.edgekey.net CNAME e3913.c
190	40.064001	192.168.1.4	23.196.96.159	TCP	34926 > http [SYN] Seq=0 win=64240 Len=0 MSS=1460 WS=8 SACK_PERM=1
191	40.136819	23.196.96.159	192.168.1.4	TCP	http > 34926 [SYN, ACK] Seq=0 Ack=1 win=64240 Len=0 MSS=1432 SACK_PERM=1 WS=7
192	40.136947	192.168.1.4	23.196.96.159	TCP	34926 > http [ACK] Seq=1 Ack=1 win=131584 Len=0
193	40.137694	192.168.1.4	23.196.96.159	HTTP	GET /DigicertGlobalRootCA.crl HTTP/1.1
194	40.211173	23.196.96.159	192.168.1.4	TCP	http > 34926 [ACK] Seq=1 Ack=255 win=64128 Len=0
195	40.212535	23.196.96.159	192.168.1.4	HTTP	HTTP/1.1 304 Not Modified
196	40.252593	192.168.1.4	23.196.96.159	TCP	34926 > http [ACK] Seq=255 Ack=462 win=131072 Len=0

Frame 1: 90 bytes on wire (720 bits), 90 bytes captured (720 bits)

Ethernet II, Src: 30:24:32:52:e2:98 (30:24:32:52:e2:98), Dst: 20:e8:82:b8:a8:76 (20:e8:82:b8:a8:76)

Internet Protocol, Src: 192.168.1.4 (192.168.1.4), Dst: 52.111.231.21 (52.111.231.21)

Transmission Control Protocol, Src Port: 34910 (34910), Dst Port: https (443), Seq: 1, Ack: 1, Len: 36

Secure Socket Layer

0000 20 e8 82 b8 a8 76 30 24 32 52 e2 98 08 00 45 00v0\$ 2R...E.
0010 00 4c 4d 3f 40 00 80 06 d0 3b c0 a8 01 04 34 6f .LM?&... ..40
0020 e7 15 88 5e 01 bb 5b 29 6a 72 bd e4 03 79 50 18 ...^..[] jr...yP.
0030 03 fb d5 c0 00 00 17 03 03 00 1f 8a 3b dd b5 7aZ
0040 80 1b a1 84 67 41 6d d0 30 d0 2b d1 de 24 04 b9gAm. 0+...\$..
0050 05 5e df f0 b4 76 0b 06 e1 c6V... ..

Intel(R) Dual Band Wireless-AC 8265 #2: <liv... Packets: 356 Displayed: 356 Marked: 0 Profile: Default

Capturing from Intel(R) Dual Band Wireless-AC 8265 #2 - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: TLSv1 Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
608	186.295643	192.168.1.4	172.217.19.35	TLSv1.2	Application Data
609	186.311991	172.217.19.35	192.168.1.4	TLSv1.2	Application Data
610	186.311991	172.217.19.35	192.168.1.4	TLSv1.2	Application Data
611	186.311991	172.217.19.35	192.168.1.4	TLSv1.2	Application Data
612	186.312248	192.168.1.4	172.217.19.35	TCP	34932 > https [ACK] Seq=1291 Ack=5493 win=129792 Len=0
613	186.314492	192.168.1.4	172.217.19.35	TLSv1.2	Application Data, Application Data
614	186.353546	172.217.19.35	192.168.1.4	TCP	https > 34932 [ACK] Seq=5493 Ack=1291 win=268800 Len=0
615	186.367976	172.217.19.35	192.168.1.4	TCP	https > 34932 [ACK] Seq=5493 Ack=1365 win=268800 Len=0
616	189.132955	192.168.1.4	192.168.1.1	DNS	Standard query A activity.windows.com
617	189.156685	192.168.1.1	192.168.1.4	DNS	Standard query response CNAME activity-consumer.trafficmanager.net A 13.68.233.9
618	189.159636	192.168.1.4	13.68.233.9	TCP	34933 > https [SYN] Seq=0 win=64240 Len=0 MSS=1460 WS=8 SACK_PERM=1
619	189.159996	192.168.1.4	13.68.233.9	TCP	34934 > https [SYN] Seq=0 win=64240 Len=0 MSS=1460 WS=8 SACK_PERM=1

Frame 611: 93 bytes on wire (744 bits), 93 bytes captured (744 bits)

Ethernet II, Src: 20:e8:82:b8:a8:76 (20:e8:82:b8:a8:76), Dst: 30:24:32:52:e2:98 (30:24:32:52:e2:98)

Internet Protocol, Src: 172.217.19.35 (172.217.19.35), Dst: 192.168.1.4 (192.168.1.4)

Transmission Control Protocol, Src Port: https (443), Dst Port: https (443), Seq: 5454, Ack: 1260, Len: 39

Secure Socket Layer

TLSv1.2 Record Layer: Application Data Protocol: http

Content Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 34

Encrypted Application Data: 91439a1a0a2292fd2d6b541107c9c7b9f02897b469394fef...

0000 30 24 32 52 e2 98 20 e8 82 b8 a8 76 08 00 45 00 0\$2R... ..v...E.
0010 00 4f ab ec 00 00 7a 06 13 14 ac d9 13 23 c0 a8 .O....z.#..
0020 01 04 01 bb 88 74 85 ee 7d 85 f2 6a 70 ec 50 18t. }.jp.P..
0030 04 1a 0f 86 00 00 17 03 03 00 22 91 43 9a 1a 0aC..
0040 22 92 fd 2d 6b 54 11 07 c9 c7 b9 f0 28 97 b4 69 ".-kT... ..C..
0050 39 4f ef d7 72 bd fc f9 89 bd 4e b8 21 90..r... ..N..

Frame (frame), 93 bytes Packets: 1208 Displayed: 1208 Marked: 0 Profile: Default

Capturing from Intel(R) Dual Band Wireless-AC 8265 #2 - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: arp Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
31	5.140094	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4
32	5.141651	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	192.168.1.1 is at 20:e8:82:b8:a8:76
88	10.337601	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4
89	10.338809	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	192.168.1.1 is at 20:e8:82:b8:a8:76
197	41.473824	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	who has 192.168.1.4? Tell 192.168.1.1
198	41.473841	30:24:32:52:e2:98	20:e8:82:b8:a8:76	ARP	192.168.1.4 is at 30:24:32:52:e2:98
211	57.467882	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4
212	57.470942	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	192.168.1.1 is at 20:e8:82:b8:a8:76
228	62.688872	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4
229	62.690191	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	192.168.1.1 is at 20:e8:82:b8:a8:76
259	78.331620	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4
260	78.332907	20:e8:82:b8:a8:76	30:24:32:52:e2:98	ARP	192.168.1.1 is at 20:e8:82:b8:a8:76
262	82.582760	30:24:32:52:e2:98	Broadcast	ARP	who has 192.168.1.1? Tell 192.168.1.4

Ethernet II, Src: 20:e8:82:b8:a8:76 (20:e8:82:b8:a8:76), Dst: 30:24:32:52:e2:98 (30:24:32:52:e2:98)

Address Resolution Protocol (request)

Hardware type: Ethernet (0x0001)
Protocol type: IP (0x0800)
Hardware size: 6
Protocol size: 4
opcode: request (0x0001)
[Is gratuitous: False]
Sender MAC address: 20:e8:82:b8:a8:76 (20:e8:82:b8:a8:76)
Sender IP address: 192.168.1.1 (192.168.1.1)
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)
Target IP address: 192.168.1.4 (192.168.1.4)

0000 30 24 32 52 e2 98 20 e8 82 b8 a8 76 08 06 00 01 05 2R... ..v....
0010 08 00 06 04 00 01 20 e8 82 b8 a8 76 c0 a8 01 01V.....
0020 00 00 00 00 00 00 c0 a8 01 04j.....

Frame (frame), 42 bytes Packets: 1528 Displayed: 34 Marked: 0 Profile: Default

Capturing from Intel(R) Dual Band Wireless-AC 8265 #2 - Wireshark

File Edit View Go Capture Analyze Statistics Telephony Tools Help

Filter: ip.addr == 192.168.1.4 Expression... Clear Apply

No.	Time	Source	Destination	Protocol	Info
1	0.000000	43.154.254.18	192.168.1.4	UDP	Source port: irdmi Destination port: 58883
2	0.329222	192.168.1.4	192.168.1.1	DNS	Standard query unknown (65) accounts.youtube.com
3	0.329481	192.168.1.4	192.168.1.1	DNS	Standard query A accounts.youtube.com
4	0.353596	192.168.1.1	192.168.1.4	DNS	Standard query response CNAME www3.l.google.com
5	0.354846	192.168.1.1	192.168.1.4	DNS	Standard query response CNAME www3.l.google.com A 172.217.19.142
6	0.356267	192.168.1.4	172.217.19.142	TCP	34985 > https [SYN] Seq=0 win=64240 Len=0 MSS=1460 WS=8 SACK_PERM=1
7	0.410540	172.217.19.142	192.168.1.4	TCP	https > 34985 [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1412 SACK_PERM=1 WS=8
8	0.410641	192.168.1.4	172.217.19.142	TCP	34985 > https [ACK] Seq=1 Ack=1 win=131072 Len=0
9	0.413501	192.168.1.4	172.217.19.142	TLSv1.2	Client Hello
10	0.431796	192.168.1.4	172.217.19.142	TCP	34986 > https [SYN] Seq=0 win=65535 Len=0 MSS=1460 WS=1 SACK_PERM=1
11	0.470898	172.217.19.142	192.168.1.4	TCP	https > 34985 [ACK] Seq=1 Ack=518 win=269312 Len=0
12	0.486887	172.217.19.142	192.168.1.4	TLSv1.2	Server Hello, change cipher spec
13	0.486887	172.217.19.142	192.168.1.4	TCP	TCP segment of a reassembled packet

Ethernet II, Src: 30:24:32:52:e2:98 (30:24:32:52:e2:98), Dst: 20:e8:82:b8:a8:76 (20:e8:82:b8:a8:76)

Internet Protocol, Src: 192.168.1.4 (192.168.1.4), Dst: 172.217.19.142 (172.217.19.142)

Transmission Control Protocol, Src Port: 34985 (34985), Dst Port: https (443), Seq: 0, Len: 0

Source port: 34985 (34985)
Destination port: https (443)
[Stream index: 2]
Sequence number: 0 (relative sequence number)
Header length: 32 bytes

Flags: 0x02 (SYN)
window size: 64240
Checksum: 0x6a8f [validation disabled]
options: (12 bytes)

0000 20 e8 82 b8 a8 76 30 24 32 52 e2 98 08 00 45 00v0\$ 2R....E..
0010 00 34 33 e1 40 00 80 06 44 cf c0 a8 01 04 ac d943...D.....
0020 13 8e 88 a9 01 bb 80 ab 7c 6c 00 00 00 00 80 02|.....
0030 fa f0 6a 8f 00 00 02 04 05 b4 01 03 03 08 01 01j.....
0040 04 02

Observed Protocols:

- TCP: Reliable transport protocol — used in web traffic and applications.
- HTTP: Application layer protocol — shows GET requests (web browsing).
- TLSv1.2: Secure connection protocol (used by HTTPS).
- DNS: Name resolution protocol (domain names to IPs).
- ARP: Resolves IP to MAC address on the local network.
- NBNS (NetBIOS Name Service): Used in Windows networks for name resolution.
- ICMP: Internet Control Message Protocol — used for diagnostics (ping).

Notable Packet Captures:

- TCP Handshakes: Multiple sequences of SYN, SYN-ACK, and ACK packets were captured, confirming the setup of reliable TCP sessions.
- HTTP Traffic: Captured GET requests to `/generate\_204`, typically used by Android or Chrome devices to check internet connectivity.
- TLS Sessions: Encrypted sessions observed between local devices and external services (e.g., Google).
- DNS Queries: Name resolution requests for hostnames like `clients3.google.com` were present, indicating normal browsing activity.
- ARP Activity: Consisted of standard "Who has" and "is-at" messages — common in local LAN activity.
- ICMP Packets: Indicated typical diagnostic operations (e.g., ping) with no anomalies.

Security & Behavior Insights:

- No evidence of malicious activities like ARP spoofing, DNS poisoning, or unauthorized access attempts.
- All TLS and HTTP traffic patterns appeared consistent with regular user browsing behavior.
- ICMP packets observed were limited and matched standard diagnostic behavior.

Protocol History:

- 100% of captured packets were Ethernet over IPv4, with all packets using the TCP protocol.
- 23.7% of the TCP traffic was secured using TLS (Transport Layer Security) — indicating encrypted communication.
- Most traffic (48.1% by volume) was TLS-encrypted, suggesting communication with secure services (e.g., HTTPS sites or secure apps).

Summary of Wi-Fi Traffic:

- Filters applied included `tcp`, `http`, `ip.addr == 192.168.1.4`, and `arp`.
- Common traffic flows were identified and categorized by protocol.
- Security checks confirmed that traffic was benign and aligned with normal operations.
- Observed patterns indicate a healthy, active local network with connections to trusted external services.

Limitations:

- Could only analyze packets to/from the host device.
- No visibility into other devices' communication due to lack of monitor mode.

5. Wi-Fi Password Strength Assessment

Methodology:

- Evaluated the network's security configuration through the router admin panel and passive network monitoring, focusing on the authentication and encryption mechanisms in place.
- Assessed the potential resilience of the Wi-Fi passphrase against unauthorized access attempts based on industry best practices.

Findings:

- The router (ZTE ZKHN H108N V2.5) is configured with WPA/WPA2-PSK authentication and TKIP+AES encryption, as observed in the admin panel.
- The passphrase field supports 8–64 characters, with the current value redacted for security.
- **Analysis:**
 - ❖ The use of WPA/WPA2-PSK with TKIP+AES indicates a mixed encryption approach. TKIP, associated with WPA, is outdated and vulnerable to attacks such as KRACK (CVE-2017-13077), while AES, used with WPA2, provides stronger protection. Transitioning to AES-only encryption is advised to mitigate these risks.
 - ❖ The passphrase length and complexity were not directly tested due to hardware constraints; however, a robust passphrase of at least 20 random characters is recommended to withstand brute-force attacks.
 - ❖ No open or WEP networks were detected during passive scans, suggesting a controlled environment.
- The assessment indicates that while the current configuration offers a baseline level of security, enhancements are necessary to align with modern standards.

Recommendations:

- ✓ Upgrade to WPA2-PSK with AES-only encryption or, if supported, implement WPA3 for enhanced security.
- ✓ Ensure the passphrase is a minimum of 20 random characters, incorporating a mix of letters, numbers, and special characters to maximize resistance to brute-force attempts.
- ✓ Consider acquiring an external USB Wi-Fi adapter (e.g., Alfa AWUS036ACH) with monitor mode support for future comprehensive testing.

Summary of Findings:

Area	Status	Notes
Device Identification	Successful	Identified three devices: router, Galaxy Tab, Realme-5 Mobile Phone, Windows PC.
Router Access	Successful	Admin panel accessible; HTTP access and TKIP encryption detected.
Port Scanning	High-risk	Router: HTTP, UPnP; Windows: SMB, RPC, AMT; potential for exploitation.
Packet Capture	Partial Visibility	Device-level traffic only; no malicious activity detected.
Password Strength Test	Assessed	Evaluated via admin panel; TKIP vulnerability noted; passphrase strength recommended.

Identified Weaknesses:

- **Open HTTP Port (80) on Router:** Allows non-secure access, risking credential interception.
- **UPnP Port (52869) on Router:** Vulnerable to device discovery and DDoS attacks.
- **Outdated Router Firmware:** Linux kernel 2.4 and last updated in 2018, susceptible to known exploits (e.g., CVE-2003-0462).
- **WPA/WPA2-PSK with TKIP:** TKIP encryption is vulnerable to attacks like KRACK (CVE-2017-13077).
- **SMB and RPC Ports (135, 445) on 192.168.1.4:** High risk of exploits like EternalBlue (CVE-2017-0144).
- **VMware Ports (902, 912) on 192.168.1.4:** Potential for unauthorized access if not needed.
- **Intel AMT Port (19992) on 192.168.1.4:** Risk of remote exploitation if misconfigured (e.g., CVE-2017-5689).
- **No visible use of segmented guest networks,** meaning all connected devices share the same internal LAN.

Recommendations:

- ✓ **Disable HTTP Access:** Configure the router to use HTTPS only for admin panel access.
- ✓ **Disable UPnP:** Turn off UPnP (port 52869) on the router to prevent device discovery and DDoS attacks.
- ✓ **Update Router Firmware:** Upgrade to a modern firmware version supporting WPA3; if unavailable, replace the router with a newer model.
- ✓ **Enforce WPA2-PSK with AES:** Disable TKIP and use AES-only encryption, or upgrade to WPA3 if supported.
- ✓ **Use a Strong Passphrase:** Ensure the Wi-Fi passphrase is at least 20 random characters to resist brute-force attacks.
- ✓ **Disable SMBv1 and Restrict Ports 135, 445:** On 192.168.1.4, disable SMBv1 and block these ports via the Windows firewall unless required.
- ✓ **Disable VMware and Intel AMT:** Close ports 902, 912, and 19992 on 192.168.1.4 unless explicitly needed for VMware or remote management.
- ✓ **Regular Updates and Audits:** Update Windows 10 on 192.168.1.4 continuously to the latest version and perform monthly Nmap scans to detect new vulnerabilities.

Conclusion:

This Wi-Fi security assessment reveals a network with significant vulnerabilities. The router's outdated firmware, HTTP access, and UPnP services pose immediate risks, while SMB and RPC ports on 192.168.1.4 are susceptible to known exploits. The use of TKIP encryption further weakens the network's security. However, no malicious activity was detected in traffic analysis, and the Galaxy Tab (192.168.1.3) and Realme-5 Mobile phone (192.168.1.2) appear secure. By implementing the recommended measures—disabling HTTP and UPnP, upgrading to WPA3 with AES, and segmenting the network—the network's security can be significantly enhanced.